

Analytic scattering model with many closed channels

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We present results from an N channel finite square well scattering model where only one channel is open. By using spatially independent couplings are short range, the scattering is nearly analytic and the density of states can be tuned. This allows for direct comparison between RMT and scattering results. Such models has important implications for molecular scattering.

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I. INTRODUCTION

I have tried to say everything that is needed to setup the scattering problem. It is rather dense...

Study simple model and see what happens when one has many closed channels. This model can be used gain an understanding of highly resonant scattering.

One more important addition is the two length scale model. This will strongly mix the bound state's character.

Then compare to RMT theory.

II. THE MODEL

This is a simple N square well model with N channels only one of which is open. This is a generalization of the 2 channel square well from Ref. [PHYSICAL REVIEW A, 65, 053617].

An interesting feature to this model is that there is an eigenbasis at short range. What this means is that the DOS/bound states will not be strongly coupled. Their "coupling" only occurs at R where they are closed. So they spectrum this model generates, might not have the properties we want. A way to over come this is to make the model a multi-step square well, or add another length scale to the problem. This way there will another step that totally scrambles the states and therefore should make the DOS or bound states spectrum strongly coupled.

It is composed of kinetic and potential energy terms

$$[T_{ij} + V_{ij}(r)]\psi_j(r) = E\psi_i(r) \quad (1)$$

Where $T_{ij} = -\frac{\hbar^2}{2m}\nabla^2\delta_{ij}$ is the kinetic energy operator and $i = \{c, o\}$ denotes the set of closed channels and the open channel. T is diagonal in open/closed channel space, this enforced with Kronecker delta. We will pick the units so that $\hbar$ is 1 and m is set to 1 (this can be changed easily).

The parameters that define the model are R , the width of the potential well; C_{ij} , the coupling between channels; D_i , the depths of the potential wells; Δ_i , the closed threshold energy; and E , the scattering energy

and wavenumber $k_o = \sqrt{2mE}$. We are able to tune the density of states with good chooses of R , C_{ij} and D_j .

The interaction for $r < R$ is

$$\begin{aligned} V_{ii}(r) &= -D_i \\ V_{ij}(r) &= C_{ij}. \end{aligned} \quad (2)$$

For $r > R$, it is

$$\begin{aligned} V_{ii}(r) &= \Delta_i\delta_{ic} \\ V_{ij} &= 0. \end{aligned} \quad (3)$$

Here δ_{ic} is 1 when i is a closed channel and zero otherwise. For all cased considered C_{ij} is symmetric and real.

We can pick the orbital angular momentum of this problem and add a centrifugal barrier simply by changing the matching functions. This leads to another parameters: L . The barrier height is $L(L+1)/2mR^2$ and this can be tuned.

The interaction matrix, V_{ij} , is an $N \times N$ matrix, which is diagonalized. From this we get the eigenvalues: ϵ_α . Furthermore, the eigenvectors can be used to form a frame transformation matrix, $U_{i\alpha}$, that defines how one transforms between the eigenbasis (α) to the open/closed basis (i).

The wavefunctions for $r < R$ are Riccati-Bessel functions (f_α) with wave number $k_\alpha = \sqrt{2m\epsilon_\alpha}$. Then the wavefunction is $\phi_{i\alpha}(r) = U_{i\alpha}f_\alpha(r)$ for the α solution.

First we will setup the general equations before reformatting them to be solved. To correctly solve the full problem, we must take a linear combination of the inside wavefunctions so that they smoothly match to both the decaying wavefunctions outside of R and have the desired scattering boundary conditions for the open channel. For $L = 0$, the equations we need to solve for a closed channel (c) are $\sum_\alpha b_\alpha\phi_{c\alpha}(R) = a_ce^{-k_cR}$ and $\sum_\alpha b_\alpha\phi'_{c\alpha}(R) = -a_ck_ce^{-k_cR}$ where $k_c = \sqrt{2m(\Delta_c - E)}$. Here b_α are the coefficients denoting the linear combination of the inside wavefunctions and a_c is the amplitude of the closed channel.

In matrix form, the equations are:

$$\Phi_{m\alpha}b_\alpha = \Psi_{mj}a_j. \quad (4)$$

$\Phi_{m\alpha}$ is $2N \times N$ matrix made up of all N channels plus N radial derivates for each channel. $\Phi_{m\alpha}$

is made up of $\phi_{i\alpha}(R)$ and $\phi'_{i\alpha}(R)$. m is an index the encompasses both i and the radial derivative and therefore index m ranges 1 and $2N$. α covers all N eigenstates. So a column of $\Phi_{m\alpha}$ is $\{\phi_{1\alpha}(R), \phi'_{1\alpha}(R), \phi_{2\alpha}(R), \phi'_{2\alpha}(R), \dots, \phi_{N\alpha}(R), \phi'_{N\alpha}(R)\}^T$.

Ψ_{mj} is a matrix the contains the correct physics behavior outside of R . It is an $2N \times (N+1)$ matrix for an N channel problem with one open channel. There is an extra coefficient for the open channel scattering boundary condition. For the closed channels, $\Psi_{mc} = g_c$ where g_c is the decaying function for c^{th} closed channel. For the open channel, we have $\Psi_{mo} = cf_o - sg_o$ and $\Psi'_{mo+1} = cf'_o - sg'_o$ where f_o (g_o) is the regular (irregular) scattering function that are the Riccati-Bessel functions. c and s combine to make $s/c = \tan(\delta)$ where δ is the scattering phase shift. So a column of Ψ_{m1} for the first closed channel is $(g_1(R), g'_1(R), 0, \dots, 0, 0)^T$. For the open channel, we have $\Psi_{m,N}$ is $\{0, 0, \dots, 0, f_o(R), f'_o(R)\}^T$ and $\Psi_{m,N+1}$ is $\{0, 0, \dots, 0, g_o(R), g'_o(R)\}^T$. Note, we have picked the open channel to the N^{th} channel.

The unknowns are b_α and a_j . b_α are the coefficients that tell us how to combine the inside wavefuctions to solve the desired boundary conditions specified by Ψ_j . a_j contains $N-1$ closed channels (a_c) amplitudes plus the scattering coefficients (c and s) for a total of $N+1$ unknowns. To solve this problem, we must add one more equation so that we have $2N+1$ equations for the $2N+1$ unknowns: b_α and a_j ($=\{a_c, s, c\}$). We typically take $a_1 = 1$, but this choice is arbitrary. Alternately, one could just let $c = 1$ and s would become the $\tan(\delta)$.

This can now be set up as standard $Ax = b$ linear solve. Picking one coefficient to be one, makes the b vector to have one element that is 1 and the rest are zero. A is $2N+1$ by $2N+1$ matrix. It is composed of $\Phi_{m\alpha}$ for the upper left block of A . The upper right block is $-\Psi_{mj}$, and this forms all but the last row of the matrix. The last row of A is simply all zero except for one element set to one, e.g. picking $a_1=1$ and our indexing scheme, we have $A_{2N+1,N+1} = 1$. So now one solves for x , it is $\{b_\alpha, a_c, c, s\}^T$.

After solving the problem, we can renormalize the final wavefunction so that the scattering amplitude is 1 or energy normalized? $c^2 + s^2 = 1$. Energy normalization is done to reflect the density of states of the scattering wavefunctions.

With our notation the full wavefunction can be written as: for $r < R$

$$\psi_i(r) = \sum_{\alpha} b_{\alpha} \phi_{i\alpha}(r) \quad (5)$$

here $i = \{c, o\}$ and $\phi_{i\alpha} = \sum_{\beta} U_{i\beta} [\delta_{\beta\alpha} f_{\alpha}(r)]$. For $r > R$ the wavefunction is:

$$\begin{aligned} \psi_c(r) &= a_c g_c(r) \\ \psi_o(r) &= cf_o(r) - sg_o(r) \end{aligned} \quad (6)$$

Here c is a index for all closed channels.

One important measure is the closed channel amplitude:

$$Z = \sum_c \int_0^\infty |\psi_c(r)|^2 dr. \quad (7)$$

The scattering cross section is

$$\sigma = \frac{4\pi}{k^2} \sin^2(\delta) \quad (8)$$

We now discuss some standard cases we study in this work for the short range interaction.

Tri-diagonal case: we take all D_i be the same for all channels (D). For the simplest case, we take C_{ij} to be constant (C) and non-zero when $j = i \pm 1$, where j and i are the channel indices. For the long range interaction we need the scattering energy ϵ and Δ_i (also taken to be constant for all channels).

GOE case: C_{ij} is defined by gaussian orthogonal ensemble. It is a bunch of random numbers, but real and symmetric. We usually take all channels to have one depth as well (D).

Perturbatively coupled open/closed case: This is where we solve a general closed channel system and In this case there is a general C_{ij} matrix, and for the closed-closed sub-block we can solve the complete bound state problem with $N-1$ closed channels. We can then perturbatively couple the known bound, closed channel system to the open channel by adding on λC_{oc} where o and c denote all the open and closed channels and λ is a parameter that can be tuned from 0 to 1. This allows us to tune a known density of resonances explicitly. It also then allows us to see how coupling to an open channel influences multi-channel molecular bound states.

III. TWO LENGTH SCALE

we can introduce another radius S and it has the same basic properties of the first well, but they are different. So like different depths and off-diagonal channel couplings. Now the eigenbasis is given by matrix V (instead of U). This scrambles the quantum number numbers and mixes the states furthers. much like rotational mixing or some other longer range interaction.

To solve this problem one would simply take the previous solution at short range $\Phi_{\alpha}(R)$ and use the new basis transformation to project this the wavefunction onto the new eigenfunctions at R . This projection then establishes the boundary condition for each channel (and each α) to match to the new basis functions. Then we can a propagate the wavefuction in each channel to S . At this point we recover the exact same problem that we had before, but with a new $\Phi_{m\alpha}$.

FIG. 1: The cross section

IV. DOS ESTIMATES

Once we have the eigenfunctions in the inner region, we are able to find the bound-states if we assume all channels terminate at R (infinite square well). Then we

are able easily estimate the resonances. We can solve the full problem by correctly searching.... While not correct, modifications are only slight. it is reasonable for large Δ . It offers a very good estimate of density of states.

V. ENERGY DEPENDENCE